

Impact of Hybrid Manufacturing and Con- formal Cooling Technologies on Injection Molding Efficiency and Component Quality

**A Comprehensive Analysis of Recent Developments and Industrial
Applications**

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Abstract:

The convergence of hybrid manufacturing methodologies—integrating computer numerical control machining with metal additive manufacturing—has fundamentally transformed injection molding tooling capabilities [1, 6, 23]. This transformation particularly manifests through the implementation of conformal cooling channels that precisely follow component geometries [7, 10]. This comprehensive investigation examines contemporary advancements in hybrid manufacturing strategies for injection mold tooling, emphasizing the substantial influence of additively manufactured conformal cooling systems on production efficiency and manufactured component quality [12, 13].

Through systematic evaluation of research published from 2023 forward, this analysis demonstrates that conformal cooling channels fabricated via hybrid manufacturing approaches achieve cycle time reductions spanning 10–40%, enhance temperature distribution uniformity by 15–40%, and substantially diminish manufacturing defects including warpage, dimensional shrinkage, and surface imperfections [4, 8, 11, 18]. The research encompasses various metal additive manufacturing technologies, including laser powder bed fusion and directed energy deposition, examining their integration with conventional machining operations, computational simulation methodologies utilizing Moldflow and ANSYS platforms, and economic viability for industrial implementation [22, 9, 10].

Findings indicate that hybrid manufacturing strategies deliver optimal equilibrium between fabrication velocity, economic efficiency, and thermal performance, particularly benefiting high-volume manufacturing scenarios [6, 17]. Future research trajectories include artificial intelligence-enhanced design optimization, topology-optimized cooling architectures, and advanced lattice-based thermal management geometries [12, 14, 20].

Key Words:

- Computer Numerical Control (CNC)
- Metal Injection Molding (MIM)
- Laser Powder Bed Fusion (L-PBF)
- Directed Energy Deposition (DED)
- Hardness Vickers (HV)
- Triply Periodic Minimal Surface (TPMS)
- Body-Centered Cubic (BCC)
- Analysis System (ANSYS)
- Computer-Aided Design (CAD)
- Not Applicable (N/A)

1. Introduction:

1.1. Initial Situation and Problem Definition:

Injection molding constitutes a cornerstone manufacturing technology for mass production of polymeric components across diverse industrial sectors, including automotive, aerospace, consumer products, and medical device manufacturing [5, 21]. The fundamental process involves injecting molten polymer material into precision-engineered mold cavities, where thermal energy extraction enables material solidification into final component geometry [7]. Despite offering exceptional advantages regarding production velocity, process repeatability, and geometric complexity capabilities, the thermal management phase presents a critical performance bottleneck, typically consuming 50–80% of total production cycle duration [5, 21].

Conventional thermal management systems employ straight-drilled cooling passages machined directly through mold core and cavity structures [7, 10]. These traditional cooling channels, constrained by subtractive manufacturing limitations, necessarily follow mold construction geometry rather than conforming to component contours [7]. This geometric limitation creates non-uniform thermal extraction patterns, generating localized thermal accumulation zones, extended solidification periods, thermal stress concentrations, and component defects including warpage, differential dimensional shrinkage, sink mark formation, and dimensional inaccuracies [7, 18]. For components featuring complex geometries with variable wall thicknesses or substantial cavity depths, conventional cooling approaches prove particularly inadequate [7, 15].

The emergence of metal additive manufacturing technologies has facilitated a fundamental paradigm transformation toward conformal cooling channel designs that maintain consistent proximity to molded component surfaces independent of geometric complexity [1, 7, 10]. Conformal cooling architecture precisely follows component contours, ensuring uniform thermal energy extraction throughout entire component geometries [7, 14]. Contemporary research demonstrates that properly engineered conformal cooling channels achieve cycle time reductions of 10–40%, improve temperature distribution uniformity up to 40%, and substantially minimize component defects [4, 8, 13, 18].

However, exclusively additive manufacturing approaches for complete mold fabrication present significant challenges, including elevated fabrication costs, extended build durations for large-scale components, inferior surface finish quality compared to precision-machined surfaces, and potential material property variations [1, 6, 10]. These limitations have catalyzed development of hybrid manufacturing strategies combining additive manufacturing's

geometric freedom with traditional computer numerical control machining's precision, surface quality, and cost-effectiveness [6, 22, 23].

1.2. Research Questions and Objectives:

This investigation provides comprehensive examination of hybrid manufacturing approaches for injection mold tooling incorporating conformal cooling technologies, analyzing contemporary developments from 2023 onwards [1, 6, 23]. The research addresses the following primary questions:

How do hybrid manufacturing techniques combine additive manufacturing with conventional machining influence injection molding cycle time reduction and component quality improvement through conformal cooling channel implementation?

The descriptive objective of this investigation encompasses comprehensive documentation of current hybrid manufacturing methodologies, additive manufacturing technologies employed for mold fabrication, and conformal cooling channel design principles established through recent research [1, 7, 12].

The analytical objective focuses on systematic evaluation of conformal cooling performance impacts on cycle time reduction, temperature distribution uniformity, and component quality improvements through quantitative analysis of recent case studies and experimental investigations [4, 8, 11, 18].

The pragmatic objective establishes practical guidelines for industrial implementation, including economic feasibility assessment, technology selection criteria, and optimization strategies for manufacturers considering conformal cooling adoption [6, 21, 23].

1.3. Methodology:

This investigation employs systematic literature research and comprehensive analysis methodologies to examine recent developments in hybrid manufacturing and conformal cooling technologies for injection molding applications [1, 7, 12].

1.3.1. Literature Research Approach:

- Database sources: Web of Science, Scopus, IEEE Xplore, ScienceDirect, Google Scholar, and industry publications [1, 6, 7, 21]
- Search terms: “conformal cooling channels,” “hybrid manufacturing,” “additive manufacturing injection molds,” “laser powder bed fusion tooling,” “directed energy deposition,” “cycle time reduction” [1, 6, 10, 22]

- Temporal scope: Primary focus on publications from 2023–2025, with foundational research from 2020–2022 [4, 8, 11, 12]
- Selection criteria: Peer-reviewed journal articles, conference proceedings from reputable sources, industry case studies from established manufacturers, citation count, author expertise, and publication venue reputation [6, 7, 21, 23]

1.3.2. Analysis Methodology:

Identified literature underwent systematic content analysis focusing on quantitative performance metrics (cycle time reduction percentages, temperature uniformity improvements, defect reduction rates), technical methodologies (manufacturing processes, simulation approaches, experimental validation techniques), and economic considerations (cost-benefit analyses, return on investment calculations, industrial feasibility assessments) [4, 8, 11, 18, 21]. Comparative evaluation across multiple studies enabled identification of consistent performance patterns and establishment of evidence-based conclusions regarding hybrid manufacturing effectiveness for conformal cooling applications [1, 6, 7, 12, 23].

1.3.3. Structure of the Work

Presents the theoretical fundamentals of conformal cooling technologies and hybrid manufacturing approaches relevant to injection mold tooling [1, 6, 7, 12]. Chapter 3 analyzes the impact of conformal cooling on cycle time, thermal performance, and component quality based on recent research findings [4, 8, 13, 18]. Chapter 4 discusses economic considerations and provides practical implementation recommendations for industrial application [6, 21, 23]. Chapter 5 summarizes the main results and critically reflects the methodological approach [1, 6, 7].

2. Relevant Fundamentals of Conformal Cooling and Hybrid Manufacturing

2.1. Evolution of Conformal Cooling Technologies:

The foundational principle underlying conformal cooling technology centers on maintaining consistent thermal resistance between cooling medium and component surface throughout the molding process [7, 10, 12]. Traditional straight-drilled cooling passages, constrained by construction geometry requirements, generate non-uniform

cooling pathways with variable distances to different component regions [7, 18]. Conformal cooling channels address this fundamental limitation by conforming precisely to component geometry, maintaining equidistant positioning throughout the thermal extraction process [7, 14, 15].

Recent developments demonstrate substantial progress in conformal cooling channel design methodologies [4, 7, 12]. Comparative simulation studies examining various conformal cooling designs across different mold materials have shown that conformal cooling achieved cycle time reductions of 43% compared to conventional channel configurations [4, 8]. These analyses revealed that conformal cooling channels not only enhance production efficiency but also reduce per-component energy consumption, contributing to more sustainable manufacturing practices [4, 14].

Applications in metal injection molding represent a previously underexplored area compared to plastic injection molding [3, 10]. Research in this domain has demonstrated that conformal cooling channels reduced mold preheating duration by 92.8% and improved cavity temperature uniformity by 15.46%, with core temperature uniformity enhancing by 3.49% [3, 8]. These improvements prove crucial for controlling green part quality, which is strongly influenced by mold temperature distribution in metal injection molding processes [3, 10].

An innovative research direction involves adaptive conformal cooling utilizing triply periodic minimal surface structures [14, 20]. Recent developments include micro-cellular cooling structures embedded within mold cores rather than traditional channel-type configurations [14, 20]. This methodology employs specific geometric configurations including Schwarz-diamond and gyroid structures to provide localized-yet-uniform cooling near thermal accumulation regions [14, 20]. This approach achieved a 40% cooling time reduction, decreasing from 25 seconds to 15 seconds compared to conventional cooling while maintaining dimensional tolerances [14, 20].

2.2. Metal Additive Manufacturing Technologies for Mold Production

Metal additive manufacturing encompasses several technologies suitable for mold fabrication, with laser powder bed fusion and directed energy deposition representing the primary technological approaches [1, 6, 10, 22]. Each technology offers distinct advantages and limitations for conformal cooling channel fabrication, necessitating careful selection based on application requirements [1, 6, 10].

2.2.1. Laser Powder Bed Fusion (L-PBF):

Also designated as selective laser melting, this technology remains the most prevalent additive manufacturing approach for mold inserts incorporating conformal cooling channels [1, 10]. L-PBF offers high dimensional resolution, excellent surface quality characteristics, and capability to produce complex internal geometries including intricate cooling channel networks [1, 10, 11]. Manufacturing demonstrations have shown that L-PBF-manufactured mold inserts with dynamic conformal cooling achieved 24% cycle time reduction, decreasing from 17 seconds to 13 seconds compared to conventional cooling approaches [13, 10]. Advanced implementations include sensorised metal additive manufacturing injection mold tools, enabling in-process monitoring of cooling performance and validation of simulation predictions [11, 10].

2.2.2. Directed Energy Deposition (DED):

This technology presents advantages including higher material deposition rates, reduced manufacturing costs, and integrated machining capabilities for achieving precise tolerances and surface finishes [1, 6]. However, directed energy deposition faces limitations in producing circular channel profiles typically required for injection molding due to stricter constraints on overhanging geometry manufacturability [6, 22]. Research has addressed this challenge by proposing self-supporting non-circular cooling channel profiles optimized for directed energy deposition manufacture [22]. This work introduced shape correcting factors to convert thermal performance calculations for non-circular profiles to equivalent circular channels, simplifying the manufacturing process while maintaining cooling effectiveness [22].

2.2.3. Material Selection Considerations:

Material selection for additively manufactured mold inserts critically impacts performance and durability characteristics [10, 19]. Maraging steel 1.2709 has emerged as a preferred material for laser powder bed fusion mold fabrication due to its excellent mechanical properties, thermal conductivity characteristics, and resistance to thermal fatigue [10, 19]. Optimized selective laser melting parameters for maraging steel have achieved sample density of 99.5%, surface roughness of 7.753 μm , and hardness of 382.8 HV, demonstrating the maturity of additive manufacturing technologies for production-quality mold inserts [10, 19].

2.3. Hybrid Manufacturing Integration Strategies

Hybrid manufacturing integrates additive and subtractive manufacturing processes to leverage complementary strengths of each technological approach [1, 6, 23]. For injection mold fabrication, hybrid strategies typically involve additively manufacturing mold inserts containing conformal cooling channels, followed by precision machining operations to achieve required tolerances, surface finishes, and dimensional accuracy for molding surfaces [6, 23].

Comprehensive hybrid manufacturing approaches for compression molding tooling using directed energy deposition with integrated machining have been demonstrated [6, 22]. These methodologies involve designing optimized conformal cooling layouts, depositing stainless steel using blown powder directed energy deposition, performing intermediate machining during deposition to ensure dimensional accuracy, and executing final finishing operations [6, 22]. This approach achieved 13.8% weight reduction compared to traditional tooling while incorporating complex conformal cooling geometries impossible to machine conventionally [6, 22].

Validation of hybrid mold inserts combining milled lower sections with laser powder bed fusion-manufactured upper sections containing conformal cooling channels has confirmed effectiveness of multi-material approaches [9, 10, 19]. Two-material strategies utilizing maraging steel 1.2709 for additively manufactured portions and tool steel 1.2343 for machined components have been validated through thermal imaging, confirming that simplified simulation models accurately predicted cooling performance with experimental results matching simulation predictions within acceptable tolerances [9, 10]. This finding suggests that complex multi-material hybrid molds can be effectively designed using streamlined computational approaches [9, 10, 19].

2.4. Computational Simulation Methodologies

Computational simulation serves as the cornerstone of conformal cooling design and optimization processes [4, 7, 12]. Three primary software platforms dominate injection molding simulation: Autodesk Moldflow, Moldex3D, and ANSYS, each offering distinct capabilities suited to different analysis requirements [4, 8, 12].

2.4.1. Autodesk Moldflow:

Provides comprehensive injection molding analysis including filling, packing, warpage, and cooling simulations [4, 7, 8]. Recent software versions have introduced automated conformal cooling channel optimization capabilities, which generate three-dimensionally printable cooling layouts based on existing circuit designs and thermal analysis results [4, 12]. This automated tool identifies regions requiring enhanced cooling and proposes optimized channel geometries to minimize cycle time and improve temperature uniformity [4, 12, 16].

2.4.2. Moldex3D:

Excels at handling complex component geometries and detailed melt behavior analysis [4, 7, 8]. Its advanced three-dimensional numerical simulation capabilities enable accurate predictions of melt front progression, cooling efficiency, and potential defect formation [4, 7, 12]. The software's comprehensive material database supports analysis of various thermoplastics, elastomers, and filled polymers [4, 8]. Integration with multi-physics analysis enables studying interactions between thermal and mechanical properties during the molding cycle [4, 12].

2.4.3. ANSYS:

Offers broader engineering simulation capabilities with emphasis on thermal and structural analysis beyond the injection molding process itself [4, 8]. ANSYS Workbench and Mechanical modules enable detailed three-dimensional transient heat transfer simulations for mold cooling analysis [8, 12]. The platform's finite element analysis capabilities support mechanical performance evaluation and stress analysis of mold structures [8, 12]. Compatibility with other ANSYS modules facilitates multi-disciplinary simulations incorporating computational fluid dynamics for detailed coolant flow analysis [8, 12].

2.4.4. Simulation workflows approaches:

Computer-aided design model preparation and meshing, material property definition and process parameter specification, filling and packing analysis, cooling simulation with channel geometry variation, warpage and shrinkage prediction, and design optimization based on analytical results [4, 7, 12, 16]. For conformal cooling applications, iterative optimization varies channel parameters while monitoring objectives including cooling time, temperature uniformity, warpage magnitude, and volumetric shrinkage characteristics [4, 8, 12, 16].

3. Analysis of Conformal Cooling Performance and Quality Improvements:

3.1.Cycle Time Reduction Performance Analysis:

Comprehensive analysis of recent empirical studies reveals that conformal cooling channels consistently deliver substantial cycle time reductions across diverse applications [4, 7, 12, 16]. The magnitude of improvement depends on component geometry complexity, material properties, and cooling system design optimization effectiveness [4, 7, 8, 12].

Table 1: Cycle Time Reduction Achieved Using Conformal Cooling:

Application Type	Cycle Time Reduction	Cooling Time Change
Simple geometry components	10–20%	13.8% improvement [4, 7, 8, 6, 22]
Complex geometry (door handle)	30%	9.6% reduction [4, 8, 16]
High-volume plastic components	40%	50% productivity increase [4, 12, 13, 21]
TPMS-based adaptive cooling	40%	25 s to 15 s [14, 20]
Metal injection molding	92.8% preheating time	Dramatic reduction [3, 8, 10]

For relatively simple geometric configurations, conformal cooling achieves 10–20% cycle time reduction compared to conventional straight-drilled channel systems [4, 7, 8]. Computational fluid dynamics simulations demonstrate faster heat extraction and more uniform temperature distributions with conformal channels in compression molding tooling applications [4, 8, 12]. The weight reduction achieved through optimized hybrid manufacturing additionally contributes to improved thermal response characteristics [6, 22].

Complex geometries featuring deep cavities or significant thickness variations exhibit more substantial improvements [4, 7, 12]. Documentation of door handle component manufacturing using conformal cooling demonstrated 30% cooling time reduction [4, 8]. The semi-annular conformal cooling channel layout reduced cooling loop pressure loss by 77% while achieving 9.6% reduction in both cycle time and cooling time, accompanied by 0.012% reduction in volumetric shrinkage rate [4, 8, 16].

High-performance applications with optimized designs demonstrate cycle time reductions exceeding 40% [4, 12, 13]. Triply periodic minimal surface-based adaptive conformal cooling structures achieved 40% cooling time reduction, decreasing from 25 seconds to 15 seconds [14, 20]. Industrial implementations report similar performance levels for high-volume plastic components, translating to approximately 50% productivity increase [4, 21].

3.2. Thermal Performance and Temperature Distribution Uniformity:

Table 2: Thermal Performance Improvements:

Performance Metric	Conventional Cooling	Conformal Cooling	Improvement
Mold temperature difference	54.939 °C	37.498 °C	31.8% [4, 8, 16]
Cavity temperature uniformity (MIM)	Baseline	Improved	15.46% [3, 8, 10]
Core temperature uniformity (MIM)	Baseline	Improved	3.49% [3, 8, 10]
Cooling performance with BCC lattices	Traditional channels	Optimized conformal	62.9% [14, 20, 16]

Quantitative assessments demonstrate substantial improvements in temperature distribution characteristics [4, 8, 16]. Optimized conformal cooling reduced mold temperature difference from 54.939°C to 37.498°C, representing a 31.8% improvement in temperature uniformity [4, 8]. This enhanced uniformity directly correlated with reduced warpage deformation, which decreased from 0.352 mm to 0.226 mm a 35.8% improvement in dimensional stability [4, 8, 18].

Advanced cooling geometries further enhance thermal performance characteristics [14, 20, 16]. Conformal cooling channels with optimized geometries including circular, serpentine, and tapered configurations incorporating body-centered cubic lattices achieved up to 62.9% better cooling performance compared to traditional channel configurations [14, 20]. The lattice structures increased effective surface area for heat exchange while maintaining structural integrity and manageable pressure drop characteristics [14, 20].

3.3.Component Quality Improvements:

Table 3: Component Quality Improvements:

Quality Metric	Improvement %	Quantitative Change
Cooling time reduction	29.35%	Optimized design [4, 7, 16]
Volumetric shrinkage reduction	5.99%	Dimensional improvement [4, 8, 18]
Warpage reduction	19.77%	Enhanced uniformity [4, 7, 18]
Sink mark depth reduction	38.85%	Surface quality [4, 7, 18]
Warpage deformation (absolute)	35.8%	0.352 mm to 0.226 mm [4, 8, 18]

Warpage reduction represents one of the most significant quality benefits [4, 7, 18]. By ensuring uniform temperature distribution, conformal cooling substantially reduces internal stresses that manifest as warpage [4, 7]. Optimal conformal cooling channel design achieved 29.35% improvement in cooling time, 5.99% reduction in volumetric shrinkage, 19.77% warpage reduction, and 38.85% reduction in sink mark depth compared to straight-drilled channel configurations [4, 7, 18].

Dimensional stability improves significantly with conformal cooling implementation [4, 7, 18]. Components exhibit better consistency in critical dimensions, tighter dimensional tolerances, and improved feature accuracy [4, 7]. Surface finish quality also benefits due to uniform cooling, reducing sink marks, flow lines, and orange peel textures [4, 7, 18].

3.4. Summary of Analysis Results

Table 4: Summary of Analysis Results:

Performance Area	Conventional Cooling	Conformal Cooling	Performance Gain
Cycle time	Baseline	10–40% reduction	Significant [4, 7, 12, 16]
Cooling uniformity	Low	High	31.8% improvement [4, 8, 16]
Warpage	High	Reduced	19.77% improvement [4, 7, 18]
Volumetric shrinkage	Baseline	Reduced	5.99% improvement [4, 8, 18]
Surface defects	Frequent	Reduced	38.85% improvement [4, 7, 18]
Energy consumption	High	Reduced	Sustainable [4, 14, 20]

Conformal cooling integrated via hybrid manufacturing consistently demonstrates measurable improvements in productivity, thermal management, and component quality [6, 12, 14, 20, 22]. Advanced designs utilizing lattice structures, triply periodic minimal surfaces, and hybrid manufacturing approaches achieve maximum performance improvements in high-volume production applications [12, 14, 20, 22].

4. Economic Considerations and Implementation Recommendations

4.1. Economic Viability Assessment

The economic viability of hybrid manufacturing with conformal cooling depends on balancing higher initial tooling investments against operational savings from reduced cycle times and improved quality characteristics [6, 21, 23]. Detailed cost-benefit analysis reveals that conformal cooling proves most advantageous for medium-to-high volume production scenarios [6, 21].

Table 5: Economic Comparison of Tooling Approaches:

Economic Factor	Conventional Tooling	Conformal Cooling Hybrid
Initial tooling cost multiplier	1.0× (baseline)	1.5× – 3.0× [6, 23]
Additive manufacturing component cost percentage	N/A	~10% of total tooling cost [6, 23]
Annual savings (5 s cycle reduction)	Baseline	\$55,000+ per mold [6, 21]
Payback period (high volume)	N/A	6–18 months [6, 21, 23]
Minimum production volume	Any volume	>100,000 parts/year recommended [6, 21]

Initial tooling costs for hybrid manufactured molds with conformal cooling typically range from 1.5× to 3× conventional mold costs [6, 23]. The additively manufactured inserts containing cooling channels represent the primary cost driver [6, 23]. However, documentation indicates that additive manufacturing tooling costs constituted only 10% of total tooling costs for an 8-cavity injection mold, suggesting that strategic use of additive manufacturing for critical cooling regions while maintaining conventional construction for other mold components optimizes the cost-performance balance [6, 23].

Cycle time reduction directly translates to operational cost savings [6, 21]. For high-volume production, even small cycle time reductions generate substantial annual savings [6, 21]. Calculations demonstrate that a 5-second cycle time reduction in high-volume production resulted in \$55,000 annual cost savings, accounting for machine time, labor, and energy consumption [6, 21]. With conformal cooling achieves 10–40% cycle time reductions, the economic benefits become compelling for production volumes exceedingly approximately 100,000 parts annually [6, 21, 23].

Quality improvements provide additional economic benefits beyond cycle time reduction [4, 6, 21]. Reduced scrap rates, lower rejection rates, and improved first-pass yield all contribute to cost savings [4, 6, 21]. The elimination or reduction of secondary operations including trimming, finishing, and inspection provides further savings [6, 21]. Energy consumption per component decreases due to shorter cycle times and more

efficient heat transfer, contributing to both cost reduction and sustainability objectives [4, 14, 20].

4.2. Practical Implementation Guidelines:

Based on comprehensive analysis of recent research and industrial implementations, the following practical recommendations emerge for manufacturers considering conformal cooling adoption:

4.2.1. Production Volume Targeting:

- Prioritize conformal cooling for high-volume production exceeding 100,000 parts annually where cycle time improvements generate rapid return on investment [6, 21, 23].
- Target complex geometries with deep cavities, varying wall thickness, or critical dimensional requirements where conventional cooling proves inadequate [4, 7, 12, 16].

4.2.2. Manufacturing Strategy Selection:

- Employ hybrid manufacturing strategies that use additive manufacturing selectively for cooling channels while maintaining conventional machining for other mold components to optimize cost-performance balance [6, 22, 23].
- Specify appropriate materials: maraging steel 1.2709 for laser powder bed fusion processes, 316L stainless steel for directed energy deposition applications, considering thermal conductivity, wear resistance, and corrosion resistance requirements [10, 19, 6, 22].

4.2.3. Design Optimization Approach:

- Invest in comprehensive simulation using Moldflow, Moldex3D, or ANSYS platforms to optimize cooling channel designs before fabrication, minimizing costly physical iterations [4, 7, 8, 12, 16].
- Implement proper coolant management including appropriate flow rates (Reynolds number 4,000–10,000), temperature control, pH monitoring, and corrosion inhibitors to maximize cooling performance and mold longevity [8, 14, 20].

4.2.4. Validation and Partnership:

- Incorporate validation testing including thermal imaging, embedded sensors, and dimensional metrology to confirm cooling performance and establish production process parameters [9, 11, 10].
- Partner with experienced additive manufacturing service providers or invest in in-house capabilities based on production volume requirements and long-term strategic objectives [6, 21, 23].
- These recommendations provide a framework for successful conformal cooling implementation, enabling manufacturers to realize documented benefits while managing costs and risks effectively [1, 4, 6, 12, 21, 22].

5. Summary and Critical Reflection:

5.1. Achievement of Research Objectives:

This investigation successfully achieved its defined descriptive, analytical, and pragmatic objectives through systematic literature research and comprehensive analysis of contemporary developments in hybrid manufacturing and conformal cooling technologies [1, 4, 6, 7, 12, 16, 22].

5.1.1. Descriptive Objective Achievement:

The research comprehensively documented current hybrid manufacturing methodologies combining numerical control machining with metal additive manufacturing technologies [1, 6, 23]. Detailed examination of laser powder bed fusion and directed energy deposition processes, their integration strategies, and conformal cooling channel design principles provided thorough theoretical foundation [10, 14, 22].

5.1.2. Analytical Objective Achievement:

Systematic evaluation of conformal cooling performance impacts revealed quantifiable improvements: cycle time reductions of 10–40%, temperature uniformity enhancements of 15–40%, warpage reductions of 19–36%, volumetric shrinkage reductions of 5–15%, and sink mark depth reductions approaching 40% [4, 7, 8, 12, 16, 18]. These metrics establish clear evidence for conformal cooling effectiveness [4, 7, 8, 12, 16, 18].

5.1.3. Pragmatic Objective Achievement:

The investigation established practical guidelines for industrial implementation, including economic feasibility assessment demonstrating payback periods of 6–18 months for high-volume production, technology selection criteria based on application requirements, and optimization strategies encompassing simulation methodologies, material selection, and coolant management approaches [6, 21, 23]. The research demonstrated that hybrid manufacturing techniques combining additive manufacturing with conventional machining substantially influence injection molding cycle time reduction and component quality improvement through conformal cooling channel implementation [6, 10, 12, 22].

5.2. Critical Reflection on Methodology:

5.2.1. Methodological Strengths:

The systematic literature research approach utilizing multiple authoritative databases ensured comprehensive coverage of recent developments [1, 4, 6, 7, 12]. Focus on peer-reviewed publications from 2023–2025 provided current, validated information [1, 4, 6, 7]. Selection criteria emphasizing source credibility enhanced reliability of analyzed sources [1, 6, 7, 21]. Comparative analysis across multiple independent studies enabled identification of consistent performance patterns [4, 7, 12, 16].

5.2.2. Methodological Limitations:

The research relied exclusively on published literature without conducting original experimental investigations [1, 4, 6, 7]. Variations in experimental methodologies, measurement techniques, and material specifications across research groups may introduce some inconsistency [4, 7, 8, 16]. The temporal scope focusing on 2023–2025 publications may have excluded some relevant foundational research from earlier periods [1, 4, 7].

5.2.3. Validity and Reliability:

Validity is supported by convergence of findings across independent research groups, consistency between simulation predictions and experimental validations, and corroboration from both academic research and industrial case studies [4, 7, 9,

11, 12, 16, 22]. Systematic search strategies and explicit selection criteria support research reliability [1, 4, 6, 7, 12].

5.2.4. Objectivity and Future Improvements:

Research-maintained objectivity through reliance on peer-reviewed publications, systematic analysis of multiple sources, and presentation of quantitative data [1, 4, 6, 7, 12]. Future investigations could enhance this research through original experimental validation, economic modeling across diverse production scenarios, extended durability trials, and exploring emerging AI-assisted design optimization and advanced lattice structures [12, 14, 16, 20].

Despite methodological limitations, the convergence of findings provides strong evidence supporting the substantial benefits of hybrid manufacturing approaches for conformal cooling implementation in injection molding applications [4, 6, 12, 22].

6. Conclusion:

This paper confirms that hybrid manufacturing combined with conformal cooling has a strong and practical impact on injection molding performance [1, 4, 6, 12, 22]. By merging the design freedom of metal additive manufacturing with the accuracy and reliability of CNC machining, manufacturers can create cooling systems that closely follow part geometry and remove heat far more efficiently than traditional straight-drilled channels [4, 7, 12, 16].

The reviewed studies consistently show that conformal cooling reduces cycle time, improves temperature uniformity, and enhances part quality by minimizing warpage, shrinkage, and surface defects [4, 7, 8, 12, 16, 18]. These benefits are most significant for complex components and high-volume production, where improved cooling directly increases productivity and lowers manufacturing costs [4, 6, 12, 21, 23].

Although hybrid tooling requires higher initial investment, the operational savings from faster cycles, reduced scrap, and lower energy usage result in relatively short payback periods [6, 21, 23]. Equally important, simulation-driven design and selective use of additive manufacturing make this approach technically reliable and economically feasible for industrial use [4, 7, 8, 12, 22].

Overall, hybrid manufacturing with conformal cooling represents a mature and scalable solution for modern injection molding, with future potential further strengthened by AI-based optimization and advanced cooling geometries [12, 14, 16, 20].

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7. Design and Fabrication of Conformal Cooling Channels in Molds: Review and Progress Updates

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8. Optimal Reynolds Number of Cooling Water in Conformal Cooling Molds

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9. Validation of Simplified Injection Molding Simulation Results for Conformal Cooling with a Hybrid Mold Insert Using Thermal Imaging Technology

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10. Metal Additive Manufacturing of Plastic Injection Molds with Conformal Cooling Channels

Authors: Baris Burak Kanbur; Yi Zhou; Suping Shen; Kim Hai Wong; Charles Chen; Abe Shocket; Fei Duan
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11. Sensorised Metal AM Injection Mould Tools for In-Process Monitoring of Cooling Performance with Conventional and Conformal Cooling Channel Designs

Authors: Mandana Kariminejad; Marion McAfee; Mohammadreza Kadivar; Christopher O'Hara; Albert Weinert; Gerard McGranaghan; Rokas Šakalys; Bruno Zluhan; Ramesh Raghavendra; David Tormey

12. Generative Design of Conformal Cooling Channels for Hybrid-Manufactured Injection Moulding Tools

Authors: Neil Wilson; Manhar Gupta; Milan Patel; Maciej Mazur; Vu Nguyen; Stefan Gulizia; Ivan Cole

The International Journal of Advanced Manufacturing Technology

DOI: <https://doi.org/10.1007/s00170-024-13754-x>

13. Dynamic Conformal Cooling Improves Injection Molding

Authors: Andreas Kirchheim; Yogeshkumar Katrodiya; Livia Zumofen; Frank Ehrig; Curdin Wick

The International Journal of Advanced Manufacturing Technology

DOI: <https://doi.org/10.1007/s00170-021-06794-0>

14. Adaptive Conformal Cooling of Injection Molds Using Additively Manufactured TPMS Structures

Authors: Seo-Hyeon Oh; Jong-Wook Ha; Keun Park

Polymers

DOI: <https://doi.org/10.3390/polym14010181>

15. Application of New Conformal Cooling Layouts to the Green Injection Molding of Complex Slender Polymeric Parts with High Dimensional Specifications

Authors: Abelardo Torres-Alba; Jorge Manuel Mercado-Colmenero; Juan de Dios Caballero-Garcia; Cristina Martin-Doñate

Polymers

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16. Optimization of Conformal Cooling Channels in Injection Molds and Investigation of SLM Molding Quality under Flow Rate Gradients

Authors: Dongyu Liu; Zeyu Yang; Zhipeng Wu; Jiajian Huang; Yuhao Zeng; Xin Shang

The International Journal of Advanced Manufacturing Technology

DOI: <https://doi.org/10.1007/s00170-025-16408-8>

- 17. Improvements in Injection Moulds Cooling and Manufacturing Efficiency Achieved by Wire Arc Additive Manufacturing Using Conformal Cooling Concept**
Authors: Alejandro Marqués; Jose Antonio Dieste; Iván Monzón; Alberto Laguía; Pascual Gracia; Carlos Javierre; Isabel Clavería; Daniel Elduque
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DOI: <https://doi.org/10.3390/polym16213057>
- 18. Warpage Optimisation on the Moulded Part with Straight Drilled and Conformal Cooling Channels Using RSM, GSO and GA Optimisation Approaches**
Authors: Mohd Hazwan Mohd Hanid; Shayfull Zamree Abd Rahim; Joanna Gondro; Safian Sharif; Mohd Mustafa Al Bakri Abdullah; Azlan Mohd Zain; Abdellah El-hadj Abdellah; Mohd Nasir Mat Saad; Jerzy J. Wysocki; Marcin Nabiałek
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DOI: <https://doi.org/10.3390/ma14061326>
- 19. Use of Maraging Steel 1.2709 for Implementing Parts of Pressure Mold Devices with Conformal Cooling System**
Authors: Jarosław Piekło; Aldona Garbacz-Klempka
Materials
DOI: <https://doi.org/10.3390/ma13235533>
- 20. Three-Dimensional Printing Conformal Cooling with Structural Lattices for Plastic Injection Molding**
Authors: Suping Shen; Baris Burak Kanbur; Chenlong Wu; Fei Duan
Frontiers in Heat and Mass Transfer
DOI: <https://doi.org/10.32604/fhmt.2024.048984>
- 21. Conformal Cooling: Higher-Quality Parts, Faster Injection Molding Cycle Times**
SyBridge Technologies Engineering Team
SyBridge Technologies (Industry article)
URL: <https://sybridge.com/conformal-cooling-higher-quality-parts-faster-injection-molding-cycle-times/>
- 22. Analysis of Self-Supporting Conformal Cooling Channels Additively Manufactured by Hybrid Directed Energy Deposition for IM Tooling**

Authors: Neil Wilson; Manhar Gupta; Maciej Mazur; Milan J. Patel; Vu Nguyen; Stefan Gulizia; Ivan S. Cole

The International Journal of Advanced Manufacturing Technology

DOI: <https://doi.org/10.1007/s00170-024-13291-7>

23. Design and Manufacture of a Cost-Effective, High-Performance Plastic Injection Mould Using a Hybrid Additive-Subtractive Manufacturing Strategy

Authors: Yuk Lun Simon Chan; Xun Xu; Olaf Diegel

Production Engineering

DOI: <https://doi.org/10.1007/s00501-025-01623-3>

8. Tools Used in This Research

This section describes the digital tools and software platforms used during the preparation, analysis, and documentation of this research. The tools supported academic writing quality, reference management, document formatting, and technical interpretation of simulation-based literature. No original experimental simulations were conducted; technical software listed below was referenced based on reviewed publications.

8.1. Academic Writing and Language Support

ChatGPT (OpenAI) was used as a writing-support tool to assist in structuring academic sections, refining technical explanations, paraphrasing complex engineering concepts, and maintaining logical consistency across the manuscript. Grammarly was employed to identify and correct grammatical errors, improve sentence clarity, and ensure an appropriate academic tone. In addition, readability-support tools such as the Hemingway Editor were occasionally used to simplify overly complex sentence structures without altering technical meaning.

8.2. Document Preparation and Formatting

Microsoft Word and LaTeX-based editors were used for document preparation, formatting, and layout management. These tools supported structured heading hierarchies, equation formatting, table alignment, figure placement, and compliance with academic formatting standards. PDF editing tools were used during the final review stage to verify layout consistency and formatting accuracy.

8.3. Reference Management and Citation Handling

Mendeley Reference Manager was used to manage bibliographic data, organize references, insert in-text citations, and generate the reference list in the required academic format. Google Scholar was utilized for literature discovery, citation tracking, and

verification of peer-reviewed sources. Cross Ref DOI services were used to validate digital object identifiers and ensure reference traceability.

8.4. Technical and Simulation Software Referenced

Although no original simulations were conducted in this study, several engineering simulation platforms were extensively referenced and critically analyzed based on reviewed literature. These included Autodesk Moldflow and Moldex3D for injection molding flow, cooling, and warpage analysis, as well as ANSYS Workbench and ANSYS Fluent for transient thermal, structural, and coolant flow simulations. These tools formed the analytical foundation for many of the performance results discussed in the reviewed studies.

8.5. Design and Manufacturing Software Referenced

Computer-aided design software such as SolidWorks, CATIA, and Creo were referenced through the analyzed literature for mold and conformal cooling channel design. Topology optimization and generative design tools were also discussed in relation to AI-assisted cooling channel development and advanced lattice-based thermal management strategies.

8.6. Data Analysis and Visualization Tools

Microsoft Excel was used to organize, compare, and summarize quantitative performance data extracted from the reviewed studies. Numerical analysis environments such as MATLAB were referenced in the literature for optimization and analytical modeling. Thermal imaging software tools were discussed in the context of experimental validation methods reported in selected studies.

9. Appendix

Appendix A: Summary Tables Used in the Study

This appendix presents the consolidated quantitative tables referenced throughout the research to support analysis transparency and result traceability.

Table 1: Cycle Time Reduction Achieved Using Conformal Cooling:

Application Type	Cycle Time Reduction	Cooling Time Change
Simple geometry components	10–20%	13.8% improvement
Complex geometry components (door handle)	~30%	9.6% reduction
High-volume plastic components	Up to 40%	~50% productivity increase
TPMS-based adaptive cooling	~40%	25 s → 15 s
Metal injection molding	92.8%	Preheating time reduction

Table 2: Thermal Performance Improvements:

Metric	Conventional Cooling	Conformal Cooling	Improvement
Mold temperature difference	54.94 °C	37.50 °C	31.8%
Cavity temperature uniformity (MIM)	Baseline	Improved	15.46%
Core temperature uniformity (MIM)	Baseline	Improved	3.49%

Lattice-based cooling efficiency	Traditional channels	Optimized conformal	Up to 62.9%
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Table 3: Component Quality Improvements:

Quality Metric	Improvement (%)	Quantitative Change
Cooling time	29.35%	Optimized design
Volumetric shrinkage	5.99%	Reduced dimensional deviation
Warpage	19.77%	Improved uniformity
Sink mark depth	38.85%	Surface quality enhancement
Absolute warpage	35.8%	0.352 mm → 0.226 mm

Table 4: Summary of Analysis Results:

Performance Area	Conventional Cooling	Conformal Cooling	Performance Gain
Cycle time	Baseline	10–40% reduction	Significant
Cooling uniformity	Low	High	31.8% improvement
Warpage	High	Reduced	19.77% improvement
Volumetric shrinkage	Baseline	Reduced	5.99% improvement
Surface defects	Frequent	Reduced	38.85% improvement
Energy consumption	High	Reduced	Sustainable

Table 5: Economic Comparison of Tooling Approaches:

Economic Factor	Conventional Tooling	Hybrid Conformal Cooling
Initial tooling cost	1.0×	1.5× – 3.0×
Additive manufacturing cost share	N/A	~10%

Annual savings (5 s cycle reduction)	Baseline	> \$55,000
Payback period	N/A	6–18 months
Recommended production volume	Any	> 100,000 parts/year